

Diode Analysis Report:

Diode: 1N4148

Manufacturer: Vishay Semiconductors

Application: Used in circuits requiring extreme fast switching

Electrical Characteristics:

Forward Voltage: 1V at 10mA

Maximum Reverse Voltage: 75V

Maximum Forward Current: 300mA

Operating Temperature Range: -65 °C to +175 °C

The maximum forward voltage of the diode is 1 V when measured at a forward current of 10 mA. It can withstand a continuous maximum reverse voltage of 75 V. The maximum forward continuous current is 300 mA. The device is designed to operate at a junction temperature up to 175 °C, while the storage temperature range is from -65 °C to +150 °C. This component is intended for extremely fast switching applications and is commonly used in high-speed digital logic circuits or as a signal processing diode where rapid response times, with a reverse recovery time of 4–8 ns, are required.